Language-Vision Model for Prompt-Driven Tumor Segmentation

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1. Introduction

Correctly identifying tumor lesions on Computed Tomography (CT) scans is a crucial task in cancer diagnosis and treatment. Developing and improving deep learning models to automate tumor segmentation could significantly reduce diagnostic errors and the time required to obtain a diagnosis. To address this issues, we designed a comprehensive pipeline for processing language–vision medical datasets and a 3D vision–language segmentation model comprising of a Language-Guided vision encoder with a Scale-aware Vision–Language Attention module, a Full-scale decoder, and a loss combining segmentation errors and contrastive errors. We compare and validate our approach on a model previously built at Gustave Roussy on abdominal CT scans with liver cancer lesions.

2. Methods for Data Processing

2.1. Dataset

The dataset used in this work is derived from the public dataset AbdomenAtlas 3.0 [3], from which we manually extracted 50 samples with liver tumor, containing each: a 3D contrast-enhanced abdominal CT scan (NiFTI [15]); the corresponding liver mask (NiFTI [15]), with voxel-level tumor segmentation; a radiology report (JSON). We limited our selection to 50 patients because identifying cases with isolated liver tumors is a time-consuming manual process.

2.2. CT Scans and Masks Preprocessing

For each patient, we processed the CT scan and its corresponding segmentation mask by using MONAI's MetaTensor [4] as core data structure, since this object combines a PyTorch tensor with associated medical imaging metadata. Preprocessing was implemented with MONAI dictionary-based transforms [18] and comprises:

• ***Reorientation*** We reoriented CT scans and masks volumes according to the radiological Right-Anterior-Superior (RAS) convention [14] to standardize the axis ordering and directions across patients and acquisition protocols.

• ***Volumes Resampling*** We resampled CT volumes and masks to isotropic voxel spacing, normalizing physical scales across patients and providing a consistent spatial frame for lesion localization. We used trilinear interpolation on CT scans and nearest-neighbor interpolation on masks to preserve discrete labels.

• ***HU Windowing and Normalization*** We clipped the Hounsfield unit (HU) values [19] stored in the CT MetaTensor voxels to a window $[hu_{min}, hu_{max}]$ that excludes outlier tissues such as dense bone or metal implants, enhancing the contrast of tissues relevant for segmentation. We then linearly mapped this window to $[0, 1]$ to obtain numerically stable inputs for training .

2.3. HU Features Extraction

We enriched our model with tissue-context information derived from HU intensities in the CT scans. For each patient, we generated a CT volume with HU values clipped to a liver–tumor window $[hu_{\min}, hu_{\max}] \approx [-150, 250]$ [1], then downsampled it by a factor $\frac{1}{s_f} = 1/16$ along each axis, to match the coarse fusion scale 3.4 and reduce memory usage. The resulting low-resolution HU volume is discretized into C_f uniformly spaced HU intervals, assigning each voxel to one interval; these indices $\in 0, \ldots, C-1$ are then converted into a C_f-channel one-hot tensor of shape $[1, C_f, d, h, w]$, where each channel encodes a specific HU interval. We use $C_f = 8$ intervals as a compact yet expressive encoding of liver and tumor tissues, providing long-range tissue context at low cost and improving sample efficiency and robustness on our small dataset.

2.4. Patch-sampling, Augmentation and Class Balance

Since training on full 3D CT volumes is unsustainable due to memory constraints, we extracted a fixed number p of 3D patches per patient. With probability P_{ratio}, we sampled the patch center from a randomly chosen tumor voxel (from the mask), yielding "tumor-centered" patches; otherwise, we sampled a random background voxel. Given a target patch size $[p_D, p_H, p_W]$, for each sampled location, we extracted the corresponding patch from the CT scan, the mask, and the HU feature volume, ensuring multimodal spatial alignment. During training , we applied simple geometric augmentations (independent random flips along each spatial axis) to CT, mask, and HU features to increase variability. We set the patch size to $[64, 64, 64]$, as a compromise between GPU feasibility and retaining sufficient contextual information around the lesion, and a $P_{\text{ratio}} = 0.8$ to compensate the imbalance between non-tumor regions and lesions.

2.5. Radiology Prompts Preprocessing

For each patient, we extracted text features from the corresponding radiology report using the Bio-ClinicalBERT tokenizer [2], obtaining an integer tensor of token indices $\in \mathbb{R}^T$ and a corresponding binary attention mask. These two tensors are then passed as input to the text encoder of our model.

2.6. Sliding Window Inference

We use sliding-window inference because the full 3D CT scan is too large to run through the model at once. Instead, the scan is processed in smaller overlapping patches, and the predictions are combined to produce a full-volume segmentation during inference.

2.7. MedVLSet Dataset Summary

Table 1 summarizes the preprocessed inputs used by our model. HU feature patches are aligned in resolution by choosing their size as $f_D \approx p_D/s_{\text{f}}, f_H \approx p_H/s_{\text{f}}, f_W \approx p_W/s_{\text{f}}$.

Modality	Dimensions	#Elements
CT patch	$[1, p_D, p_H, p_W]$	p
Mask patch	$[1, p_D, p_H, p_W]$	p
HU features patch	$[C_f, f_D, f_H, f_W]$	p
Prompt token IDs	$[T]$	1
Attention mask	$[T]$	1

Table 1: Preprocessed inputs per training sample.

3. Proposed Model

3.1. Text and Vision Embedding

• ***Text Embedding*** Token IDs and attention masks are fed to the frozen Bio-ClinicalBERT encoder [2], which returns token embeddings, then linearly projected to $\mathbb{R}^{B\times T\times C_{\text{txt}}}$ to match our architecture. We set $C_{\text{txt}} = 256$ as a compromise that preserves sufficient expressivity while remaining computationally efficient and consistent with the visual embedding size. Tokens $L_{tok} \in \mathbb{R}^{B\times T\times C_{\text{txt}}}$ and mask $M \in \mathbb{R}^{B\times T}$ are then sent to model's encoder.

• ***Visual Embedding*** The raw CT patch is downsampled and mapped to a higher-dimensional feature space $V \in \mathbb{R}^{B\times C_{\text{in}}\times p_D\times p_H/4\times p_W/4}$ by a 3D convolution. We chose $C_{\text{in}} = 64$ and this stride-4 downsampling as a compromise that preserves sufficient spatial detail while keeping the 3D encoder we built computationally tractable on patch-based training.

3.2. Language-Guided Vision Encoder (LGVE)

To facilitate deep interaction between linguistic features and the complex visual environment of medical images, we designed a 3D adaptation of a Language-Vision Encoder developed in previous work [16], that leverages a language-guided attention module.

In LGVE we used Instance Normalization [21] to train with a small batch size without noise, making features less sensitive to global intensity shifts between patients and acquisitions. The other layers in LGVE and the four stages, that produce the multimodal feature maps $F_i \in \mathbb{R}^{C_{vi}, p_D, p_{H_i}, p_{W_i}}$ from the first embeddings V and L_{tok} and that build our encoder, mirror the original reference. In our 3D adaptation, the downsampling block [5] uses convolutions with stride $(1, 2, 2)$, since clinical semantics along CT depth are sensitive to collapse. The chosen configuration of channels and LGVE blocks is summarized in Table **??**. It enforces progressive channel growth to match feature complexity (fewer channels at higher resolution, more at coarser, more semantic scales) and concentrates LGVE blocks where semantic fusion is most critical and the language–attention module has the greatest impact. Alternative configurations with similar principles are plausible and worth exploring experimentally. A full mathematical explanation of our 3D implementation of the encoder is provided in the appendix. 9.3

Stage	Spatial resolution	Channels	#LGVE blocks
F_1	$p_D \times \frac{p_H}{4} \times \frac{p_W}{4}$	64	3
F_2	$p_D \times \frac{p_H}{8} \times \frac{p_W}{8}$	128	3
F_3	$p_D \times \frac{p_H}{16} \times \frac{p_W}{16}$	320	5
F_4	$p_D \times \frac{p_H}{32} \times \frac{p_W}{32}$	512	2

Table 2: Multi-stage 3D LGVE encoder: spatial resolutions, channels and number of LGVE blocks per stage.

3.3. Scale-Aware Vision-Language Attention (SVLA)

We implemented a 3D adaptation of the Scale-Aware Vision–Language Attention module introduced in previous work [16], and we briefly outline here the main differences from the original 2D design. Compared to the 2D SVLA in the original paper, all convolutions, norms and attention reshaping are generalized to 3D, and directional kernels are explicitly anisotropic along depth and in-plane axes to account for volumetric CT structure, in particular long-range spatial context is captured by three pairs of anisotropic 3D convolutions with large kernels along (H, W), (D, W) and (D, H).

This module is the core of our encoder, as it learns scale-aware visual representations from diverse receptive fields via convolutional kernels of different sizes, while integrating linguistic features with visual information consistently across multiple scales.

3.4. Full-Scale Decoder with HU Features Fusion

Our decoder aggregates multi-scale, language-guided CT and HU features at a common 3D resolution, refines them with a dedicated interaction block, and then upsamples to produce dense voxel-level tumor logits. In detail:

• ***HU Branch*** After text and CT encoding, we fuse HU features into the deep visual stream at the F_3 scale. The HU volume $f \in \mathbb{R}^{B\times C_f\times f_D\times f_H\times f_W}$ is projected to 64 channels, to match the later fusion width, and passed through an LGVE block so it can be modulated by token-level text embeddings L_{tok}.

• ***Fusion*** Once the encoder outputs F_1, F_2, F_3, F_4, we upsample the deep CT map F_3 to the HU coarse grid and compute a pooled sentence embedding $L \in \mathbb{R}^{B\times C_{\text{txt}}}$, which is broadcast to the same 3D size. The three modalities L_{space}, F_3 and f are then concatenated along channels and reprojected to obtain $F_3^{\text{fused}} \in \mathbb{R}^{B\times c_3\times f_D\times f_H\times f_W}$, which replaces F_3 as input to the decoder.

• ***Full-Scale Decoder*** We designed a decoder that fully fuses shallow and deep features from multiple encoder stages [10]. In particular, it takes the three deepest scales (F_2, F_3^{fused}, F_4) and reconstructs a dense segmentation by upsampling to F_1 size, concatenating the upsampled maps

$$x = [P_2,\ P_3,\ P_4] \in \mathbb{R}^{B\times(c_1+c_2+c_3)\times d_1\times h_1\times w_1}$$
$$\xrightarrow{\text{Hamburger \& MLP}} \text{logits}_{\text{low}} \in \mathbb{R}^{B\times 1\times d_1\times h_1\times w_1}$$

and processing them through a 3D adaptation of the Hamburger block [8]. The final prediction is produced by a lightweight MLP head (we used again instance normalization [21] to obtain a stable training with small batch-size) on the fused feature volume, following standard practice for deep representations [7]. The produced low-resolution logits are then upsampled to match the original CT patch size: logits $\in \mathbb{R}^{B,1,p_D,p_H,p_W}$.

3.5. Loss Function

In our experiments, we use a combined loss that penalizes both segmentation error, measuring how accurately lesions are localized on the CT patch, and a contrastive error, which enforces proper alignment between text and visual embeddings.

• ***Segmentation Loss*** We define our segmentation loss as

$$\mathcal{L}_{\text{seg}} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.5\,\mathcal{L}_{\text{BCE}}.$$

We chose Dice loss $\mathcal{L}_{\text{Dice}}$ [13] since it is largely insensitive to foreground–background class imbalance, which is crucial in this setting. We complement it with a voxel-wise Binary Cross-Entropy (BCE) loss $\mathcal{L}_{\text{BCE}}$, which penalizes per-voxel classification errors and encourages well-calibrated probability outputs. We rely on a symmetric contributions from

Dice and BCE so that over- and under-segmentation are penalized in a balanced way. This is desirable in lesion segmentation, where the objective is to maximize overlap between predicted and true tumor regions under strong class imbalance, rather than to privilege false positives or false negatives a priori.

• ***Vision-Language Contrastive Loss*** To align CT and text representations in a shared embedding space, we adopt a symmetric InfoNCE-style contrastive loss [22]. Making the loss symmetric is beneficial because it enforces consistency in both retrieval directions (CT must retrieve its report and the report must retrieve its CT), reducing asymmetric solutions where one modality dominates, and yields a tighter alignment of CT–text pairs in the joint space (see Appendix 9.6 for the full implementation). We chose a temperature $\tau = 0.07$, that controls the sharpness of the softmax over similarities, to make the contrastive term numerically comparable to the segmentation loss.

The resulting combined loss has the following expression:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{contrast}}\, \mathcal{L}_{\text{contrast}},$$

We chose $\lambda_{\text{contrast}} = 0.1$ to make the contrastive term a regularizer, keeping segmentation as primary objective.

3.6. Evaluation Metrics

We used Dice Score [6] to evaluate models performance, since it is directly comparable with Dice loss and balances false positives and false negatives on the lesion region, which is what matters clinically, while accuracy or per-voxel error can be dominated by background.

4. Baseline Model

We compare our proposed architecture with a model previously developed at Gustave Roussy. This model is a language-vision model that relies on a ResNet as encoder and that fuses HU features, along with visual and text features, in a decoder built on different transposed convolutional layers. Its full implementation is described in our appendix. 9.4

This model exhibited three key limitations: incomplete end-to-end design, weak integration of textual and visual information, and a loss function that did not adequately enforce alignment between language intent and spatial predictions. As a result, segmentation outputs were often inconsistent, insufficiently guided by text, and prone to inaccurate or fragmented lesion boundaries. Our method specifically targets these shortcomings by redesigning both the architecture and training objective to better couple language and vision throughout the segmentation process. It integrates language guidance early via scale-aware attention, enforces multi-scale consistency through a full-scale decoder, and corrects training misalignment using a combined segmentation and contrastive loss. This results in more coherent, clinically relevant segmentations.

5. Experiments

5.1. Objective

Our goal is to assess whether our model is well-designed, learns effectively, and outperforms the baseline. We compare the two models in experiments that vary the number of patches per patient (one vs. multiple), trying to isolate the effect of the two architectures.

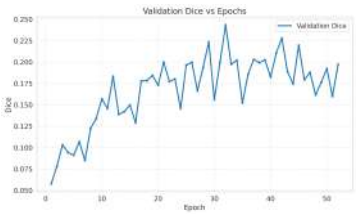
5.2. Training Settings

• We used the Adam [9] optimizer with learning rate $l_r = 10^{-4}$ and weight decay $w = 10^{-4}$ as a standard, robust optimization setting widely adopted in medical image segmentation, and we kept it fixed across all experiments to isolate the effect of architectural changes between the baseline and our model.

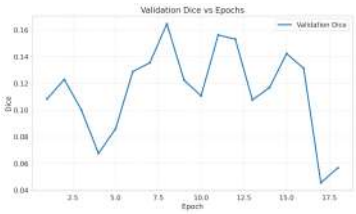
• We randomly split the dataset into 40 training patients and 10 validation patients. We trained for up to epochs $= 100$ with batch size $B = 3$ (limited by 3D CT memory cost), monitoring the mean validation Dice score at each epoch and applying early stopping with a patience of 20 epochs. We applied augmentation to the train set to improve model generalization in such small dataset. 2.4

Metric	Baseline	Proposed
Avg Train Combined Loss	0.6032	0.5975
Avg Train Segmentation Loss	0.5173	0.5289
Avg Train Contrastive Loss	0.8590	0.6852
Avg Validation Loss	0.6826	0.5911
Best Validation Dice	0.1763	0.1379

Table 3: Average training and validation metrics evaluated on the two models in Experiment 1.



(a) Dice score VS Epochs-Proposed model

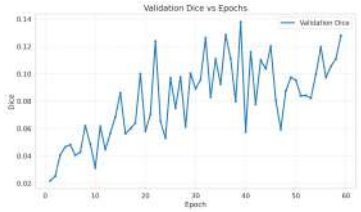


(b) Dice score VS Epochs-Baseline model

Figure 1: Dice score trend - Experiment 2.

5.3. Experiment One: Training with One Patch

We trained both models with a single 3D patch per patient ($p = 1$) for both training and validation 2.4. The resulting average losses and Dice scores over all epochs are reported in Tab. 3. Although the baseline achieves a higher max validation Dice, Fig. 2 shows a clearer upward trend for the proposed model, indicating more consistent learning. The strong oscillations in both curves stem from stochastic patch-based validation on a very small validation set, where a few "easy" or "hard" patches strongly affect the mean, and from the intrinsic instability of Dice on small lesions, where slight over/under-segmentation causes large relative changes in the score. In 3, the baseline model exhibits a decreasing training loss, but the validation loss rapidly flattens, indicating limited extraction of generalizable features and a sub-optimal inductive bias that favours overfitting. In contrast, the proposed model shows closely matched training and validation losses, confirming effective learning without evident overfitting on such a small dataset. The slightly lower validation loss is expected, since data augmentation is applied only to the training set 2.4, making validation samples comparatively easier. Since our small dataset prevents us from making more consistent evaluations, we increase p in the second experiment.

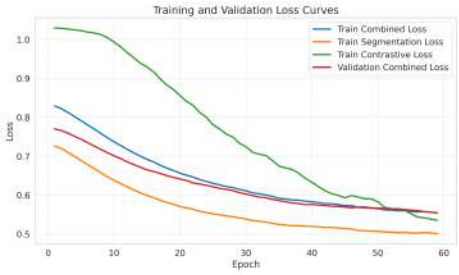


(a) Dice score VS Epochs-Proposed model

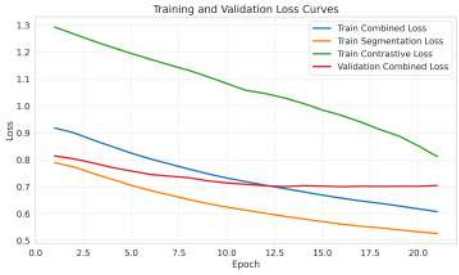


(b) Dice score VS Epochs-Baseline model

Figure 2: Dice score trend - Experiment 1.



(a) Losses VS Epochs-Proposed model



(b) Losses VS Epochs-Baseline model

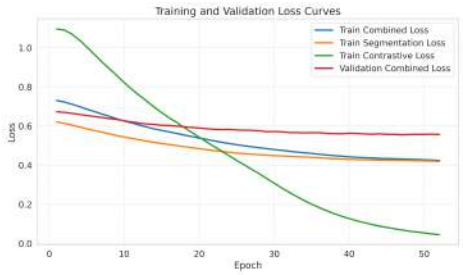
Figure 3: Losses - Experiment 1.

5.4. Experiment Two: Training with Multiple Patches

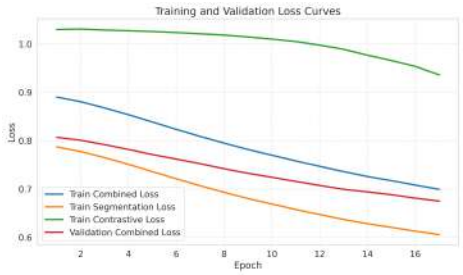
We trained both models with $p = 8$ patches per patient in the training set and $p = 4$ in the validation set, to amplify the information extracted from each case and partially compensate for the small dataset. Our model reaches a competitive Dice score given the data regime and, as shown in Fig. 1, exhibits a steadily improving Dice trajectory, whereas the baseline displays a much less consistent trend. For the proposed model, both segmentation and combined training losses 4 decrease smoothly, and the contrastive loss drops markedly, indicating that seeing more patches per patient strongly improves vision–language alignment. The validation loss first decreases and then plateaus, which is compatible with a very small, patch-based validation set (the train:val patch ratio moves from $1:4$ to $1:32$) that quickly saturates its information content, and possibly with a relatively high weight on the contrastive term that may limit further gains in validation segmentation loss. Overall the gap is stable and the model is learning, opening the door to further experiments to address the issue. The baseline behaves like a relatively low-capacity model: it benefits from more data (all losses go down) but cannot exploit the extra patches to strongly align image and text or to significantly reduce the validation loss, hence the weaker Dice.

Metric	Proposed	Baseline
Avg train combined loss	0.4730	0.6115
Avg train segmentation loss	0.4461	0.5233
Avg train contrastive loss	0.2681	0.8820
Avg validation loss	0.5701	0.6798
Best validation Dice	0.2434	0.1646

Table 4: Average training and validation metrics evaluated on the two models in Experiment 2.



(a) Loss vs. epochs – proposed model



(b) Loss vs. epochs – baseline model

Figure 4: Loss curves for Experiment 2.

6. Inference

We compare inference of 1 CT scan against the four models mentioned above (proposed and baseline training with one or multiple patches). We show the results in Appendix. Despite stable optimization behavior, favorable training and validation losses, the qualitative segmentation results ar suboptimal. This highlights the limitation that loss values alone do not fully capture spatial accuracy or clinical relevance, particularly in patch-based, class-imbalanced medical segmentation tasks. Small misalignments in patch aggregation, sensitivity to background context, and error accumulation during sliding-window inference can lead to anatomically implausible predictions despite low loss. Moreover, the inference study suggests that irrelevant tissues and large homogeneous background regions can distract the model when relying primarily on HU-based intensity features.

7. Conclusion

Through our experiments we validated the proposed architecture against the baseline, showing that it can exploit multimodal features more effectively and achieve consistently higher Dice scores. However, the overfitting observed in the second experiment and the poor generalisation on held-out patients indicate that the current setting is not sufficient. The inference study suggests that irrelevant tissues can distract the model, motivating the replacement of HU-based features with richer TotalSegmentator-derived anatomical representations [23] on a larger cohort. Lastly, rebalancing the loss (decreasing the weight of the contrastive term and increasing the Dice contribution) is a natural direction to prioritise segmentation quality.

8. Ethical Concerns

[24] One significant ethical risk identified in this project is the potential for clinical harm due to inaccurate tumor segmentation. This risk was surfaced through two key dimensions of the Digital Ethics Canvas. Firstly, 'Welfare' due to incorrect segmentation outputs. such as missed lesions or poorly defined tumor boundaries, could negatively influence diagnosis or treatment planning, leading to delayed or missed care or inappropriate interventions. It may also cause psychological harm from misdiagnosis. Secondly, 'Autonomy' due to risk of automation bias, where clinicians may place undue trust in visually compelling model outputs, even when uncertainty is high.

This risk affects patients (direct stakeholders) and clinicians (indirect stakeholders). Given the medical context, the severity of this harm is high, while the likelihood is moderate due to known limitations of deep learning models in generalizing across imaging conditions and tumor presentations.

The risk was evaluated through a review of prior work on medical image segmentation and through empirical validation. Existing literature highlights persistent challenges related to robustness and generalization in medical imaging models [11] and emphasizes that high quantitative performance does not guarantee clinical reliability [20]. Additional studies show that segmentation models often fail on small or low-contrast lesions, which are clinically critical [12]. Empirically, model performance was assessed using quantitative metrics such as the Dice coefficient, complemented by qualitative inspection of segmentation overlays. These evaluations highlighted cases where model confidence did not always align with segmentation accuracy, reinforcing the risk identified through the Canvas.

To account for this risk, the project explicitly treats the model as a research and decision-support tool, not a diagnostic system. All outputs are intended to be reviewed by a human expert, and visual inspection is emphasized alongside numerical metrics. However, mitigation was limited by constraints on expanding the training dataset. With a larger dataset, we could find more edge cases, such as small lesions and low-contrast tumors, where errors are more likely and pay more attention to them. Most importantly, it would help generate a more robust and generalizable pipeline.

9. Appendix

9.1. HU Windowing and Normalization Transformation

$$I_{i,j\,\text{tensor}} = \frac{\text{clip}(I, hu_{min}, hu_{max}) - hu_{min}}{hu_{max} - hu_{min}}$$

9.2. Text and Vision Embedding

• ***Text Embedding*** Token IDs and attention masks are fed to the frozen Bio-ClinicalBERT encoder [2], which returns token embeddings $H \in \mathbb{R}^{B\times T\times \text{hid}}$, where B is the batch size and hid is the BERT hidden dimension. These embeddings are then linearly projected to $\mathbb{R}^{B\times T\times C_{\text{txt}}}$ to match our architecture.

9.3. Encoder

LGVE block. Given a visual feature map $V \in \mathbb{R}^{B\times C\times D\times H\times W}$ and token–level text embeddings $L \in \mathbb{R}^{B\times T\times C_{\text{txt}}}$ (with mask $M \in \{0,1\}^{B\times T}$), one `LGVEBlock` performs:

$$\begin{aligned}
\tilde{V} &= \text{SVLA}(V, L, M) && \in \mathbb{R}^{B\times C\times D\times H\times W}\\
X &= \text{IN}_1(\tilde{V} + V) && \in \mathbb{R}^{B\times C\times D\times H\times W}\\
Z &= W_2\,\sigma(W_1 X) && \in \mathbb{R}^{B\times C\times D\times H\times W}\\
Y &= \text{IN}_2(Z + X) && \in \mathbb{R}^{B\times C\times D\times H\times W},
\end{aligned}$$

where:

- $\text{SVLA}(\cdot)$ is the Scale-aware Vision-Language Attention module (output form equal to V);
- IN_1, IN_2 are InstanceNorm3d on the channels;
- $W_1 : \mathbb{R}^C \to \mathbb{R}^{4C}$, $W_2 : \mathbb{R}^{4C} \to \mathbb{R}^C$ are realised with two conv $1\times1\times1$ (FFN) [7];
- σ is a non-linear GELU applied element-wise.

Then `LGVEBlock` is a residual block:

$$V \xrightarrow{\text{SVLA + res + IN + FFN + res + IN}} Y \quad (\text{same shape as } V).$$

LGVEStage3D. A `LGVEStage3D` takes as input $X^{(s-1)} \in \mathbb{R}^{B\times C_{\text{in}}\times D\times H\times W}$ and produce $F_s \in \mathbb{R}^{B\times C_{\text{out}}\times D\times H'\times W'}$ through:

1. **Downsampling / channel projection**

$$U^{(s)} = \begin{cases} \phi\big(\text{IN}(\text{Conv}^{(1,2,2)}_{3\times3\times3}(X^{(s-1)}))\big), & \text{if do_downsample = T} \\ \phi\big(\text{IN}(\text{Conv}_{1\times1\times1}(X^{(s-1)}))\big), & \text{if } C_{\text{in}} \neq C_{\text{out}}, \\ X^{(s-1)}, & \text{else,} \end{cases}$$

   where ϕ is ReLU. If `do_downsample=True`, spatial dimensions become

$$U^{(s)} \in \mathbb{R}^{B\times C_{\text{out}}\times D\times \frac{H}{2}\times \frac{W}{2}}.$$

2. **Stack of LGVE Blocks** We define $X^{(s)}_0 = U^{(s)}$ for $k = 0, \ldots, n_s - 1$:

$$X^{(s)}_{k+1} = \text{LGVEBlock}^{(s)}_k\big(X^{(s)}_k, L, M\big),$$

   where $X^{(s)}_k \in \mathbb{R}^{B\times C_{\text{out}}\times D\times H'\times W'}$ for all k.

3. **Stage Output**

$$F_s := X^{(s)}_{n_s} \in \mathbb{R}^{B\times C_{\text{out}}\times D\times H'\times W'}.$$

MultiStageLGVEEncoder3D. Multistage Encoder considers already embedded volume

$$X^{(0)} \in \mathbb{R}^{B\times C_{\text{in}}\times D\times \frac{H}{4}\times \frac{W}{4}}$$

and apply in stack all the four stages with channels $(c_1, c_2, c_3, c_4) = (64, 128, 320, 512)$:

$$\begin{aligned}
F_1 &= \mathcal{S}_1\big(X^{(0)}, L, M\big) && \in \mathbb{R}^{B\times c_1\times D\times \frac{H}{4}\times \frac{W}{4}},\\
F_2 &= \mathcal{S}_2\big(F_1, L, M\big) && \in \mathbb{R}^{B\times c_2\times D\times \frac{H}{8}\times \frac{W}{8}},\\
F_3 &= \mathcal{S}_3\big(F_2, L, M\big) && \in \mathbb{R}^{B\times c_3\times D\times \frac{H}{16}\times \frac{W}{16}},\\
F_4 &= \mathcal{S}_4\big(F_3, L, M\big) && \in \mathbb{R}^{B\times c_4\times D\times \frac{H}{32}\times \frac{W}{32}},
\end{aligned}$$

where $\mathcal{S}_s$ is the `LGVEStage3D` of stage s. In other words:

$$X^{(0)} \xrightarrow{\mathcal{S}_1} F_1 \xrightarrow{\mathcal{S}_2} F_2 \xrightarrow{\mathcal{S}_3} F_3 \xrightarrow{\mathcal{S}_4} F_4,$$

with progressively reducing spatial resolution in H, W and number of channels that grow from C_{in} until $c_4 = 512$.

9.4. Baseline Model in Detail

In this section, we describe the standard model previously built at Gustave Roussy.

- **Vision Encoder** The CT patch encoder is a MONAI's 3D ResNet-10 [17]. After four residual stages with strided convolutions and pooling, it outputs deep feature maps $f_{CT} \in \mathbf{R}^{B\times 512\times d_4\times h_4\times w_4}$ at an effective stride of 16 along each axis. The HU features are projected from 8 to 64 channels and the CT features are upsampled to their coarse grid (f_D, f_H, f_W).
- **Text Encoder** A frozen Bio-ClinicalBERT encoder processes the tokenized prompt with an attention mask, and a masked mean pooling over tokens yields a single sentence embedding in $\mathbb{R}^{256}$, after a linear projection from the original BERT hidden size 768.
- **Fusion** The projected HU features (64 ch.), the upsampled CT features (512 ch.) and the broadcast 256-channel text embedding are concatenated along the channel dimension into a fused multimodal volume $x \in \mathbb{R}^{B\times 832\times f_D\times f_H\times f_W}$.
- **Decoder** The decoder takes x and upsamples it through two 3D transposed convolutions, progressively increasing the spatial resolution to $4f_D \times 4f_H \times 4f_W$. A final trilinear interpolation to the original CT resolution (p_D, p_H, p_W) followed by a $1\times 1\times 1$ convolution produces the segmentation logits $\in \mathbb{R}^{B\times 1\times p_D\times p_H\times p_W}$.

9.5. Dice and BCE Loss

The full expression of the Dice loss is

$$\mathcal{L}_{\text{Dice}}(p, y) = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon},$$

where $p_i \in [0, 1]$ are the predicted probabilities (after sigmoid), $y_i \in 0, 1$ the ground-truth labels, and $\epsilon > 0$ is a small constant for numerical stability.
Voxel-wise Binary Cross-Entropy (BCE) loss expression is the following. Given predicted probabilities $p_i \in [0, 1]$ and ground-truth labels $y_i \in \{0, 1\}$ at voxel i, BCE is defined as

$$\mathcal{L}_{\text{BCE}}(p, y) = -\frac{1}{N}\sum_{i=1}^{N}\big[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\big],$$

where N is the number of voxels. In practice we use the "with logits" variant, which applies the sigmoid to the network logits internally and improves numerical stability.

9.6. Contrastive Loss Full Implementation

Let $V_i \in \mathbb{R}^C$ and $L_i \in \mathbb{R}^C$ be the visual and text embeddings for sample i in a mini-batch of size B, after ℓ_2-normalization. We build a similarity matrix

$$s_{ij} = \frac{V_i^\top L_j}{\tau}, \qquad 1 \le i, j \le B,$$

where $\tau > 0$ is a temperature (implemented as a learnable or fixed logit scale). Assuming aligned pairs (V_i, L_i) as positives, the image-to-text InfoNCE loss is

$$\mathcal{L}_{\text{i}\to\text{t}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(s_{ii})}{\sum_{j=1}^{B}\exp(s_{ij})},$$

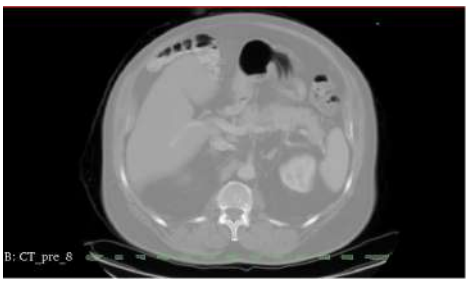
and the symmetric text-to-image loss is

$$\mathcal{L}_{\text{t}\to\text{i}} = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(s_{ii})}{\sum_{j=1}^{B}\exp(s_{ji})}.$$

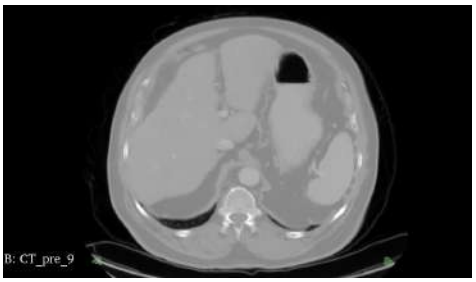
Our contrastive objective is the average of the two directions:

$$\mathcal{L}_{\text{contrast}} = \frac{1}{2}\big(\mathcal{L}_{\text{i}\to\text{t}} + \mathcal{L}_{\text{t}\to\text{i}}\big),$$

which encourages matched CT and text embeddings to have high similarity while pushing apart mismatched pairs, following the InfoNCE formulation [22].



(a) Results with Proposed Architecture and Multiple Patches



(b) Results with Proposed Architecture and Single Patch

Figure 5: Masks (green) overlaid on pre-processed CT scans (images from 3DSlicer)

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